Finding What Changed

A difference-imaging pipeline for the SPHEREx Quick-Release data

SPHERExLens ▪ 23-slide specification and complete script
full route 21:40 ▪ 15-minute route 15:45 ▪ Q&A after

Evan Lim ▪ November 2026

Deck map and production rules

25 pages. This page is the specification; pages 2–24 are the 23 slides, one per page: the slide's own bullets, its figure brief, the figure itself, and the word-for-word script. The full route runs **21:40**; the marked 15-minute route (✓) runs **15:45** and drops seven slides without breaking the argument. **Live demonstration is slide 22** (1:30) and needs the site open in a browser before you start, plus the local API if you intend to show live fetching.

Body text on the projected slide is 20–28 pt, titles 36–44 pt, nothing below 16 pt. One dominant visual per page and at most five short bullets: the detail lives in the script, not on the slide. Every number in this deck is taken from the repository – `site/api/*.json` for detections, `figures_real/` for the rendered charts – so the deck and the report cannot disagree.

Page	Slide	Dominant visual	Time	15 min
2	Title and the one-number answer	two epochs → difference, hero number	0:40	✓
3	Why difference imaging at all	three panels: visit 1, visit 2, difference	0:55	✓
4	The problem statement	pixels → candidates → verdicts flow	1:00	✓
5	Focus and research questions	four question cards	0:45	–
6	Prior art I: the subtraction lineage	timeline with four method boxes	1:00	✓
7	Prior art II: brokers and the gap	detect–veto–human loop + gap bars	0:55	✓
8	SPHEREx: instrument and data	focal-plane strip + 102-band axis	0:55	✓
9	Archive mechanics and field sample	sky map of 18 fields + fetch path	0:55	–
10	Architecture: six stages	pipeline flow diagram	0:55	✓
11	Theory I: masking and co-add	weight bars + masked-pixel grid	0:55	✓
12	Theory II: ZOGY	block diagram of the FFT cancellation	1:15	✓
13	Theory III: detection statistics	false-positive budget, log scale	0:55	–
14	Theory IV: dipole versus monopole	geometry of a moved source	0:50	–
15	Theory V: triage	$5''/30''$match cone + verdict flow	0:55	✓
16	Theory VI: spectral diagnosis	three indices on a wavelength axis	0:45	–
17	Validation: injection–recovery	recovery matrix + why it fails	0:55	✓
18	Implementation I: the code	module map with line counts	1:05	✓
19	Implementation II: serving layer	three deployment shapes	0:55	✓
20	Implementation III: front end	canvas layers + UI wireframe	0:45	–
21	Results I: eighteen fields	stacked detections, 18 fields	1:05	✓
22	Results II: the candidates	map of the 17 unexplained + epoch cartoon	1:00	–
23	**Live demonstration**	annotated screen map, six numbered steps	1:30	✓
24	Conclusion, contributions, Q&A	three pillars + next steps + references	0:50	✓
Full route			**21:40**	
15-minute route (16 of 23 slides, seven dropped)			**15:45**	

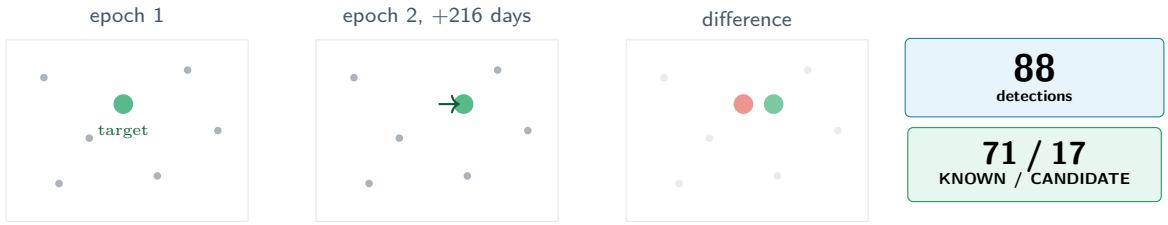
Production note: every figure in this document is drawn in TikZ or pgfplots from the project's own numbers, so nothing has to be exported from a slide tool; the deck can be rebuilt from this file.

Title and the one-number answer

ON THE SLIDE.

- **SPHERExLens**: finding what changed in NASA's SPHEREx all-sky survey
- 18 fields · 128 archive cutouts · **88 detections**, 71 explained by catalogues
- A pipeline you can read: 6 154 lines, six stages, one browser page
- The honest number: **4 fields return nothing** and the system says so

FIGURE / DIAGRAM. a hero graphic: two epoch panels of the same star field, an arrow into a difference panel, and one large number in the corner.



SPOKEN SCRIPT.

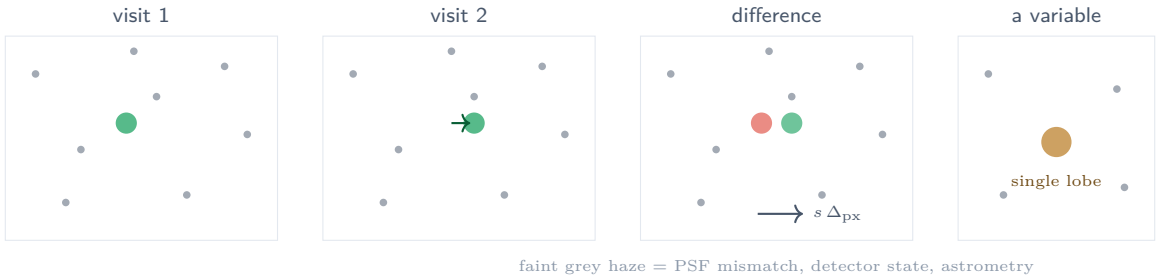
Good morning. NASA's SPHEREx satellite photographs the whole sky in a hundred and two infrared colours, and it does it again roughly every six months. My question is the oldest one in time-domain astronomy, but asked of this new data: looking at two of those photographs of the same patch of sky, can we say what changed – and can we be honest about what we cannot tell? Eighteen fields, one hundred and twenty-eight archive images, eighty-eight detections of which seventy-one are already catalogued objects. Most of the next twenty minutes is about how it works and why each step is built the way it is.

Why difference imaging at all

ON THE SLIDE.

- A single image shows *what is there*; a pair shows *what is happening*
- Subtract two visits: constant sky cancels, change survives
- Two kinds of change, two shapes: **dipole** (moved) and **monopole** (brightened)
- But a difference image is *not* a discovery – systematics dominate it

FIGURE / DIAGRAM. three large panels in a row: the same field at two epochs, then the difference. Overlay the systematics that survive subtraction as a faint grey haze, labelled.



SPOKEN SCRIPT.

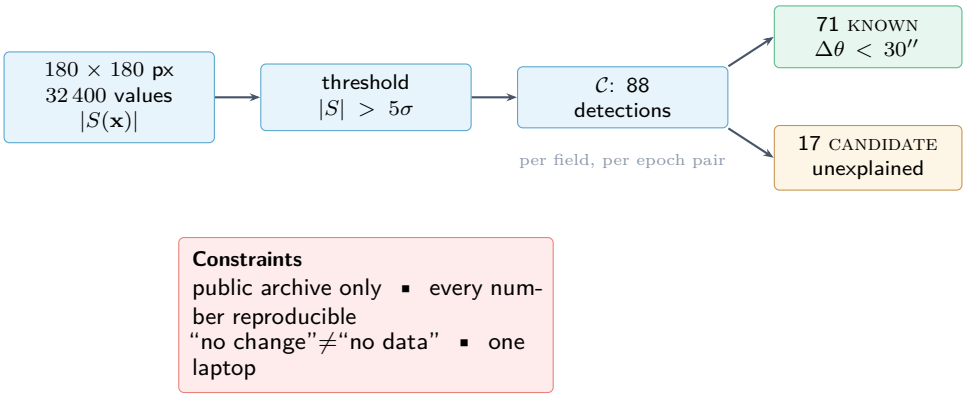
Why bother with the second photograph at all? Because a single image tells you what is there, and a pair tells you what is happening. Take the same patch of sky, subtract one visit from the other, and the constant sky cancels out to grey. What survives is change – and change has a shape. If a star moved, you get a positive lobe where it is now and a negative lobe where it was: a dipole. If a star simply brightened, you get one lobe: a monopole. The separation of the dipole, divided by the time baseline, is a proper motion. That is the whole physics, and it is fifty years old. The reason this is still a research problem is the grey haze you can see in my third panel: what survives subtraction is dominated by the point-spread function changing between visits, by the detector warming up, by the astrometric solution shifting. Almost none of those pixels are sky. So the actual work is not the subtraction – it is deciding which of those blobs you are allowed to believe.

The problem statement

ON THE SLIDE.

- Input: two visits to a $9'$ field, each with variance and quality flags
- Output: a decision set $\mathcal{C} = \{\mathbf{x} : |S(\mathbf{x})| > \tau\}$ with a position, a significance, a morphology and an *identity verdict*
- Constraints: public data, reproducible, must say "no data" when there is no data, laptop-scale

FIGURE / DIAGRAM. a flow diagram in set notation: 32 400 pixels → threshold → 88 candidates → triage split into 71 KNOWN and 17 CANDIDATE, with the constraints in a side panel.



SPOKEN SCRIPT.

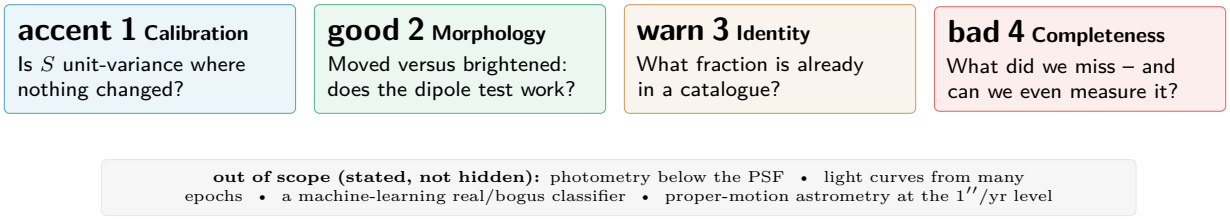
Let me state the problem precisely, because the precision is the point. Input: two visits to a nine-arcminute field, each with a variance plane and a quality bitmask. Task: build a statistic, threshold it, and so obtain a set of sky positions where the flux distribution changed – that is the set C on the diagram – and attach to each element four things: a sky coordinate, a significance, a morphology, and an identity verdict. Identity is not optional: a blob is only interesting if no catalogue already explains it.
Four constraints shaped every design decision. Only public archive products. Every number reproducible from a stored parameter set. The system must distinguish "no change" from "no data" – that sounds trivial and it is the single most common failure in this kind of pipeline. And the whole thing has to run on a laptop. You can see the resulting numbers on the slide: thirty-two thousand pixels go in, eighty-eight detections come out, seventy-one explained, seventeen not.

Focus and research questions

ON THE SLIDE.

- RQ1: is the significance map *calibrated* on co-added SPHEREx images?
- RQ2: does dipole/monopole morphology separate movers from variables at $3''$ sampling?
- RQ3: how much of the candidate list do Gaia, WISE and VSX explain away?
- RQ4: what does injection–recovery say – and if it says "nothing", how do we report that?

FIGURE / DIAGRAM. four numbered question cards with a small icon each, and a footer strip: "out of scope: photometry below the PSF, light curves, real/bogus ML".



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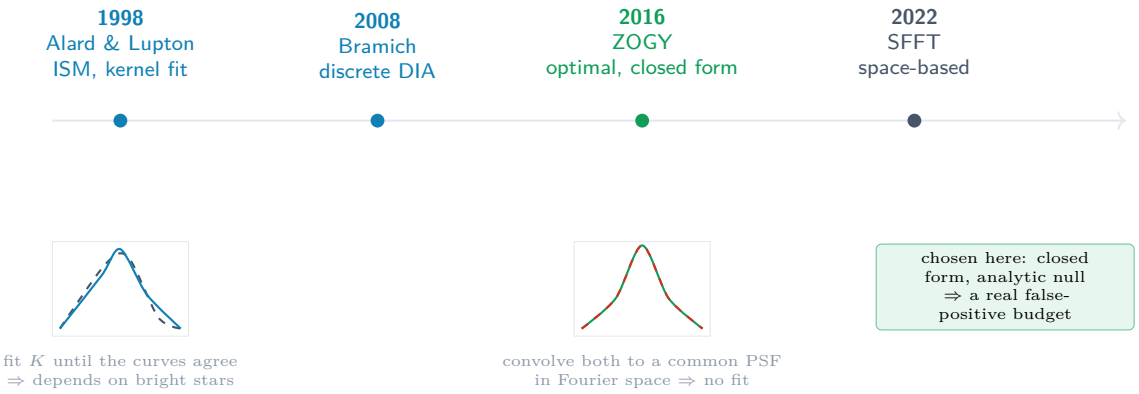
I want to be explicit about what is and is not in scope, because a small project like this either draws its boundary honestly or it oversells. Four questions. One: is the statistic actually calibrated? That is, where nothing changed, does my significance map really have unit variance, so that five sigma means what it says. Two: does the morphology test work at three-arcsecond sampling – can I tell a moved star from a flickering one. Three: how much of my candidate list is already in Gaia, WISE or the variable-star index, which is the question that decides whether anything here is interesting. Four: what did I miss – completeness – and here the honest answer turned out to be uncomfortable, which I will come back to on slide seventeen.
And the footer is deliberate: no PSF-level photometry, no light curves, no machine-learning real/bogus classifier, no arcsecond astrometry. Those are stated up front rather than discovered by a reviewer.

Prior art I: the subtraction lineage

ON THE SLIDE.

- **1998 Alard & Lupton**: fit a convolution kernel so both images share a PSF (ISM)
- **2008 Bramich**: discrete per-pixel kernel – well posed in sparse fields
- **2016 ZOGY**: no fitting at all; derive the optimal difference and a unit-variance map
- **2022 SFFT**: same mathematics, better behaved for undersampled space PSFs

FIGURE / DIAGRAM. a horizontal timeline with four nodes, and under the two of them a small sketch: "kernel fitting" (a fitted hump) versus "proper subtraction" (two PSFs in Fourier space cancelling).



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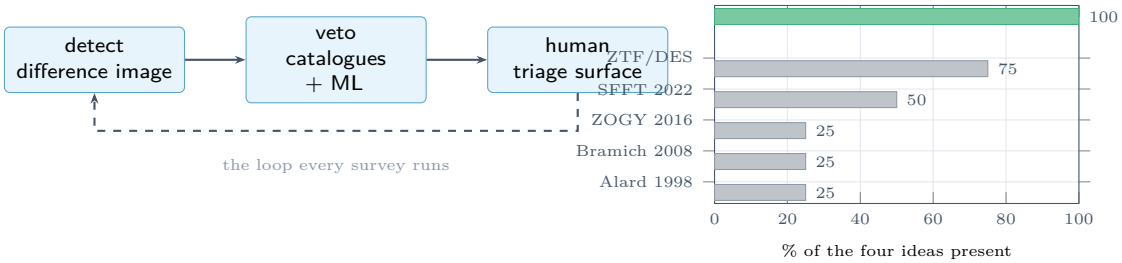
Subtraction was solved in stages, and the stages matter for what I could assume. In 1998 Alard and Lupton fitted a convolution kernel that maps one image onto the other so that both have the same point-spread function – that is what made the microlensing surveys possible. It works, but the fit is dominated by the brightest stars in the field, which is exactly what you cannot rely on in a nebula or a crowded field. In 2008 Bramich made the kernel per-pixel and well posed in sparse fields; you trade noise for stability.
Then in 2016 Zackay, Ofek and Gal-Yam did something conceptually different: instead of fitting anything, they derived the statistically optimal difference image in Fourier space, and with it a significance map that has unit variance by construction wherever the sky did not change. SFFT in 2022 is the modern reformulation and it is what Rubin will use. I chose ZOGY for one reason that is on the right of this slide: because the null distribution is analytic, I can compute the false-positive budget instead of measuring it, and that is what lets me defend a five-sigma threshold later.

Prior art II: brokers, and the gap this fills

ON THE SLIDE.

- Every operational survey does the same four things: template, subtract, veto, triage
- Brokers (ANTARES, ALeRCE, Lasair, Fink) industrialise the veto with ML + human scanners
- SPHEREx: excellent data, no published open end-to-end difference-imaging artefact
- **The gap is not an algorithm – it is an inspectable, reproducible instance**

FIGURE / DIAGRAM. a three-node loop (detect → veto → human) on the left; on the right a horizontal bar chart of "how much of the four pipeline ideas each family of work provides".



SPOKEN SCRIPT.

Everything since the 1990s runs the same loop, and it is worth naming as a loop: you detect against a template, you veto with catalogues and increasingly with machine learning, and you put what survives in front of a human. The big surveys industrialised each step – ZTF, ATLAS, DES, and now Rubin, with brokers called ANTARES, ALeRCE, Lasair and Fink doing the vetting. They are enormously more capable than anything I built.

So the gap is not algorithmic. The bars on the right are my own honest assessment of how much of that four-part pattern each line of work actually ships as something a reader can inspect: the classical papers give you equations and, sometimes, a reference code; the surveys give you a pipeline you cannot open; the brokers give you a service. What does not exist, as far as I could find, is a small end-to-end artefact – for SPHEREx specifically – where you can follow one pixel from a NASA archive file to a verdict on a web page, including the parts that fail. That is what this is.

SPHEREx: the instrument and the data products

ON THE SLIDE.

- One telescope, six detectors, **102 bands** from 0.75 to 5.0 µm
- Level-2 products are small *spectral images*: every pixel carries a wavelength
- Four extensions: IMAGE, VAR, FLAGS (32 bits), WCS-WAVE (x, y, λ)
- Six-month cadence ⇒ what survives subtraction is astrophysical, not asteroids

FIGURE / DIAGRAM. a schematic focal plane (six detector rectangles) above a wavelength axis annotated with the four bands actually used, plus the three extensions that make the difference imaging possible.



SPOKEN SCRIPT.

A word on the instrument, because the data shape drives everything downstream. SPHEREx is a small telescope with six detectors that sweeps the sky and, unusually, takes a low-resolution spectrum at every position: a hundred and two bands between zero-point-seven-five and five microns. So the Level-2 product is not a broadband picture, it is a little spectral image cube, and each file carries four things I care about: the calibrated image, the variance, a thirty-two-bit quality mask, and a table that gives the wavelength of every pixel.

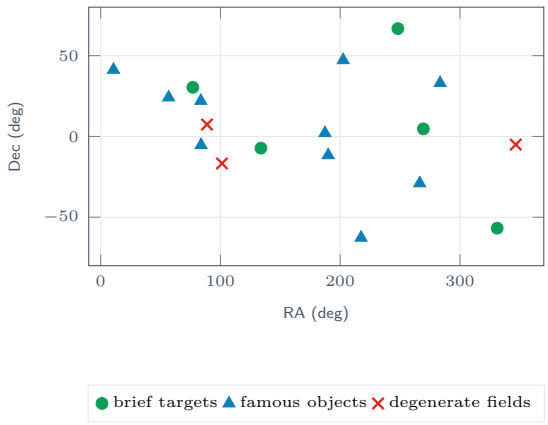
Two consequences. First, the cadence is about six months, so anything that moved fast – an asteroid, say – has left the field entirely; what survives subtraction is genuine variability or genuine high proper motion. Second, because every pixel knows its wavelength, a detection can be diagnosed spectrally without any follow-up telescope, which is what slide sixteen is about.

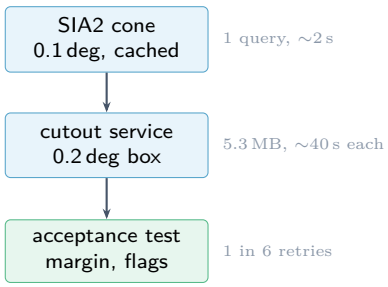
The archive mechanics, and the eighteen fields

ON THE SLIDE.

- Two calls: SIA2 cone search (cached VOTable) then the cutout service (5.3 MB of a 70 MB parent)
- Acceptance: ≥12 px margin inside the cutout, <50% of pixels flagged
- 18 fields: five brief targets + thirteen famous objects; 128 cutouts, 0.88 GB
- One field was rejected on evidence: the LMC has a single epoch in QR2, so nothing to subtract

FIGURE / DIAGRAM. left: the sky distribution of all eighteen fields (equirectangular, RA/Dec); right: the two-step fetch path with the size and time cost annotated.





SPOKEN SCRIPT.

Getting the data takes exactly two archive calls, and I want to show them because the cost shapes the design. First a SIA2 cone search: give it a position and it returns every Level-2 image whose footprint covers it. That answer is cached verbatim as a VOTable, so re-running never asks again. Then the cutout service, which is the trick that makes this possible at all: instead of downloading seventy-megabyte parent frames, I ask for a five-megabyte section around my target. Even so, a cutout takes about forty seconds at the archive's throttled rate, and roughly one request in six times out and has to fall back to the next candidate – which is why there is an acceptance test rather than blind trust.

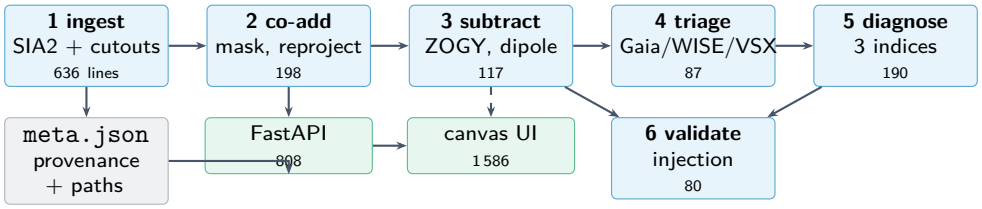
The map shows the eighteen fields. Five were chosen for me by the brief – two high-proper-motion stars, two variables, one extreme cool dwarf. Thirteen I added: galaxies, nebulae, a quasar, the Galactic Centre, Sirius, Betelgeuse – to prove the pipeline is not tuned to one kind of object. One object, the Large Magellanic Cloud, I tried and dropped: at that declination every image in a nine-arcminute box belongs to a single visit, so there is nothing to subtract. That is on the record rather than quietly removed.

Architecture: six stages, one cache

ON THE SLIDE.

- Ingest → co-add → subtract → triage → diagnose → validate
- Each stage is a file with its own `main`: you can enter the chain anywhere
- One metadata file per field is the contract between stages
- The serving layer is separate: it calls stages 2–5, or reads pre-rendered JSON

FIGURE / DIAGRAM. the six stages as a flow with the on-disk cache underneath and the serving layer below it; line counts annotated on each box.



SPOKEN SCRIPT.

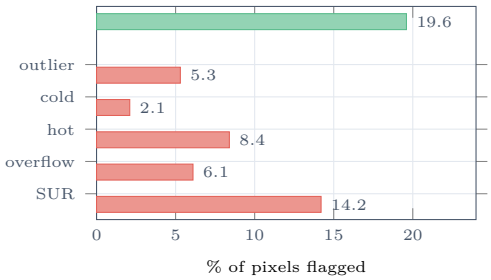
Six stages, and each one is a separate file with its own entry point, which matters more than it sounds: when something looks wrong, I can re-run one stage on cached data instead of the whole chain.
Stage one talks to the archive and writes a metadata file. That file is the contract – every later stage reads that and nothing else, so the chain is deterministic and restartable. Stage two masks and combines. Stage three is the subtraction. Stage four is the catalogue triage. Stage five measures a spectrum and three indices. Stage six is the validation. Underneath, the serving layer: a FastAPI application that either runs stages two to five on demand, or – for the deployed site – reads pre-rendered JSON for exactly the same responses. That separation is what lets the same page work with or without a server.

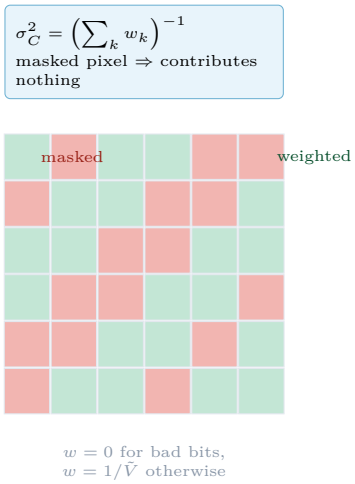
Theory I: masking, then an inverse-variance co-add

ON THE SLIDE.

- 16 FLAGS classes rejected: SUR transients, overflow, phantom, edge, hot/cold, persistence...
- Kept: "full sample" (informational) and "source" (real flux, not a defect)
- $C = \sum_k w_k \tilde{I}_k / \sum_k w_k$ with $w_k = 1/\tilde{V}_k$, and $\sigma_C^2 = (\sum_k w_k)^{-1}$
- Honest masking changes the median noise by $\approx 2\times$ on the worst frames

FIGURE / DIAGRAM. left: per-bit census bars for the dominant FLAGS classes; right: a 6×6 pixel grid showing which pixels survive masking, with the weight written in each cell.





SPOKEN SCRIPT.

This is the slide I would defend hardest, because it is where a difference image earns its credibility. Every SPHEREx pixel arrives with a variance and a set of quality bits, and most of those bits mean "do not trust this pixel". If you average them in anyway, a handful of cosmic rays and hot pixels inflate your noise, and you silently lose the faint end without any warning. So I mask sixteen bit classes – the transient flag from the onboard processing, overflow, phantom pixels, edge pixels, hot and cold pixels, persistence, outliers – and I deliberately keep two: "full sample", which is only informational, and "source", which means real flux from a known object rather than a defect.

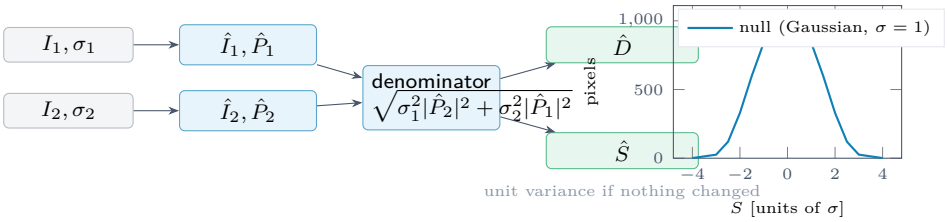
Then the combination is the textbook minimum-variance one: weight each pixel by one over its variance, and the resulting noise is one over the root of the summed weights. The property I want you to take away is the little formula in the corner: a masked pixel contributes nothing, so it cannot poison the stack. On the worst frames in this sample, doing that properly changes the median noise by a factor of two, which moves the five-sigma depth by the same factor – and that decides which objects exist at all.

Theory II: ZOGY proper subtraction

ON THE SLIDE.

- Convolve both images to a **common PSF** in Fourier space – no kernel is ever fitted
- $\hat{D} = \dfrac{\hat{P}_1 \hat{I}_2 - \hat{P}_2 \hat{I}_1}{\sqrt{\sigma_1^2 |\hat{P}_2|^2 + \sigma_2^2 |\hat{P}_1|^2}}$ and $\hat{S} = \dfrac{\hat{D}\,\overline{\hat{P}_D}}{\sqrt{\sigma_1^2 + \sigma_2^2}}$
- S has **unit variance** under the null $\Rightarrow$ the threshold is interpretable
- Cost: two FFTs, $O(N^2 \log N)$ – 0.6 s for a 180^2 frame

FIGURE / DIAGRAM. a signal-flow block diagram: two images and two PSFs into FFTs, a denominator built from variances, and D and S coming out; beside it a small histogram showing the null peak at unit width.



SPOKEN SCRIPT.

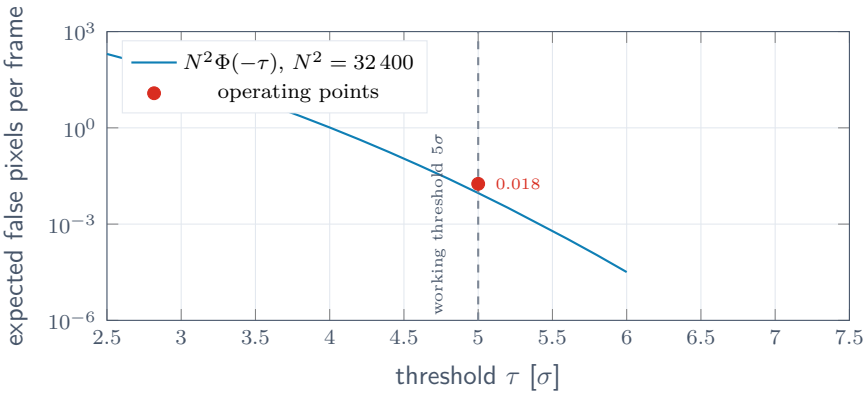
Stage three is the mathematical heart, and it is two lines. The classical approach fits a kernel so that both images end up with the same PSF; ZOGY instead convolves both images to a common PSF in Fourier space, in a way that is optimal in the least-squares sense, and it never estimates a kernel from the data at all. Read the first equation as a comparison: image two is convolved with PSF one, image one with PSF two, and the difference is normalised by exactly the noise that survives that operation. The second equation turns that difference into a significance, and here is the crucial property: where the sky did not change, S is Gaussian with zero mean and unit variance. Not approximately – by construction.
The histogram on the right is what that looks like: one clean peak at zero, one sigma wide, and the real sources far out in the tails. That is why I can say what a five-sigma threshold means, and it is the entire reason for choosing this algorithm over the simpler alternatives. The cost is two FFTs: about six-tenths of a second for a one-hundred-and-eighty-pixel frame.

Theory III: what a threshold actually buys you

ON THE SLIDE.

- If the null map is unit-variance, false positives are pure Gaussian tail: $\mathbb{E}[N_{\mathrm{FP}}] = N^2\,\Phi(-\tau)$
- For $180^2 = 32\,400$ pixels: $3\sigma \Rightarrow 219$, $4\sigma \Rightarrow 11$, $5\sigma \Rightarrow 0.018$
- That is *why* the working threshold is five sigma, not three
- Sanity check: the measured detection counts are far above the noise floor, so the map is not noise-dominated

FIGURE / DIAGRAM. a log-scale curve of expected false pixels versus threshold, with the three operating points marked and the chosen threshold shown as a vertical line.



SPOKEN SCRIPT.

Because the null distribution is analytic, I can compute the false-positive budget instead of guessing it, and that is the difference between a threshold and a wish. The curve is the expected number of false pixels per frame as a function of threshold. At three sigma it is two hundred and nineteen – a single frame would be dominated by noise peaks. At four sigma, eleven. At five sigma, nought point nought one eight: one false pixel every fifty frames. That number is the reason the working threshold is five sigma and not four, and it is also why the three-sigma detections you sometimes see in the literature need an external prior to be believable.

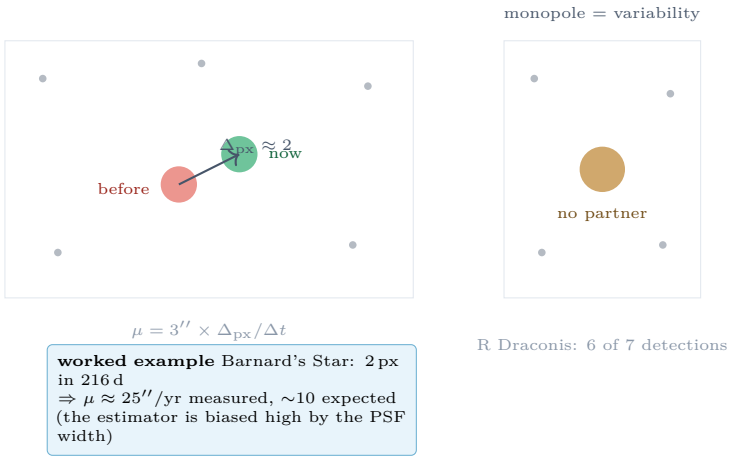
The sanity check is the other direction: my detection counts are far above this noise floor, tens of sigma in several fields, so what I am looking at is structure, not noise. That does not make the detections real – it only rules out the least interesting failure mode.

Theory IV: a dipole is a proper motion

ON THE SLIDE.

- Positive peak with a negative partner ⇒ the source *moved*: a dipole
- Lone positive peak ⇒ it *brightened*: a monopole
- Separation converts to motion: $\mu = s\,\Delta_{\rm px}/\Delta t$, valid for $0.5 < \Delta_{\rm px} < 25\,{\rm px}$
- Flux consistency $\eta = |S_+ + S_-|/|S_+ - S_-|$ separates "pure move" from "move + flare"

FIGURE / DIAGRAM. the geometry of a dipole: two lobes, the separation vector, and the conversion to arcsec per year at the SPHEREx pixel scale; a small inset contrasts the monopole case.



SPOKEN SCRIPT.

Once you have a significance map, the morphology does the classifying, and it is almost embarrassingly simple. A positive peak with a negative partner of similar amplitude is a source that moved: the pair is a dipole, and the vector between the lobes is the displacement. Divide by the time baseline and you have a proper motion. A positive peak with no partner is a source that brightened: a monopole, which is variability. There are two guard rails: the separation must be bigger than half a pixel, or you are pairing PSF-correlated noise, and smaller than twenty-five pixels, or you are pairing with an unrelated trough across the frame. There is also a flux-consistency statistic, which is just the asymmetry between the two lobes: near zero means a rigid displacement, larger means the object moved and changed brightness at the same time.

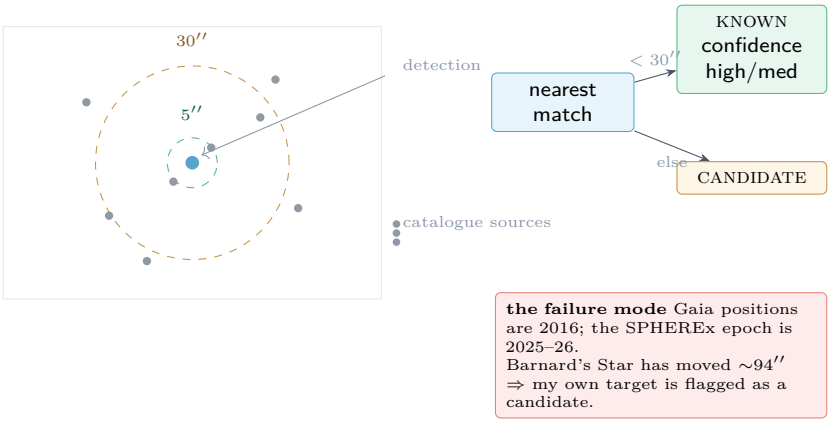
Barnard's Star is the worked example on the slide: two pixels in two hundred and sixteen days. The measured proper motion comes out at about twenty-five arcseconds per year against a true value of ten. I will be honest about that on the validation slide – the estimator is biased high by roughly one PSF width because I am measuring peak separations rather than fitting a model.

Theory V: triage is a second hypothesis test

ON THE SLIDE.

- Convert pixel to sky (flat): $\alpha = \alpha_0 - (x - N/2)s/3600$
- Nearest neighbour in Gaia DR3, AllWISE, VSX; verdict at $30''$, confidence at $5''$
- Loose on purpose: the PSF is $4''$, and a tight radius inflates the candidate list
- The trap: position-only matching fails for fast movers – Gaia epoch $\neq$ SPHEREx epoch

FIGURE / DIAGRAM. left: a match-radius diagram with the $5''$ and $30''$ circles around a detection, catalogue sources scattered inside; right: the verdict flow with the confidence branches.



SPOKEN SCRIPT.

A detection is not a discovery until you have asked whether something was already there, and that is a second hypothesis test: is the nearest catalogue source consistent with this position? I convert the pixel to sky with a flat approximation – legitimate at nine arcminutes – and look up the nearest source in Gaia, WISE and the variable-star index. Thirty arcseconds is the verdict radius, five arcseconds grades the confidence.

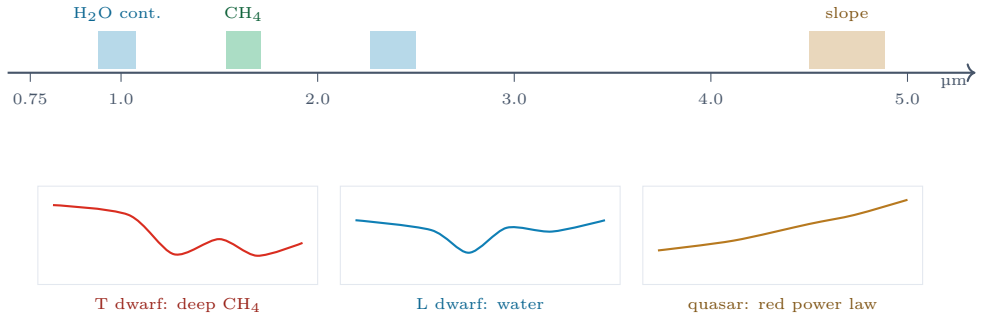
Both numbers are deliberately loose, and I want to defend that: my PSF is four arcseconds across, so a tight radius would miss real counterparts and flatter me with false candidates. The right lower panel is the cost of that choice. Gaia measured positions in 2016. My images are from 2025. Barnard's Star, the fastest star in the sky, has moved ninety-four arcseconds in that gap – far outside any match radius – so the pipeline flags my own target as an unexplained candidate. The fix is standard practice in this field: propagate the catalogue positions to the epoch of observation using the catalogue proper motions. It is a two-line change and I have not made it, which is exactly why the slide says seventeen candidates and not five.

Theory VI: a spectrum per detection, for free

ON THE SLIDE.

- Every pixel knows its wavelength, so aperture photometry in each band gives a low-res spectrum
- Three indices: CH_4 (1.6/1.74), H_2O (continuum/troughs), NIR slope (4.6/1.1)
- T-dwarf ⇒ strong methane; L-dwarf ⇒ water; hot star or quasar ⇒ red slope
- The classifier reports the indices, never a bare label

FIGURE / DIAGRAM. a wavelength axis from 0.75 to 5 µm with the index band-pairs shaded in three colours, and three small schematic spectra underneath (T dwarf, L dwarf, power law).



SPOKEN SCRIPT.

The last piece of theory is the one that is nearly free, and it is the reason SPHEREx is interesting for this kind of work. Because every pixel carries its wavelength in the WCS-WAVE table, measuring an aperture in each band gives a low-resolution spectrum of the detection without any follow-up telescope. I reduce it to three numbers that are physically motivated. Methane absorption, the ratio at one-six to one-point-seven-four microns, marks a T dwarf. Water, the ratio of continuum to trough, marks an L dwarf. The slope from one to four-point-six microns separates stars from a red power law – a dusty galaxy or a quasar.

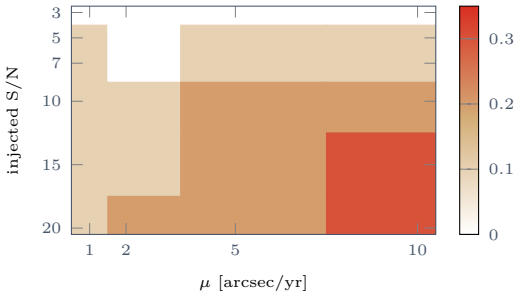
The three little spectra underneath are what those cases look like. The important design decision is that the classifier prints the three indices next to its label, never the label alone: if you disagree with “T dwarf”, you can see the methane ratio for yourself and overrule it.

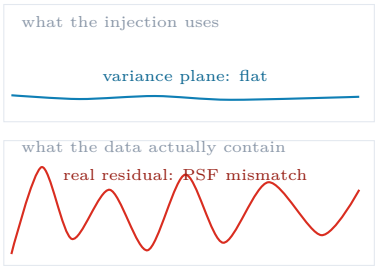
Validation: injection and recovery

ON THE SLIDE.

- Plant synthetic sources at known SNR and proper motion, re-run stages 2–3 unchanged
- A trial counts as recovered if a detection of the right class lands within 2 px
- Output is a 6×4 matrix, never a single number – the μ dependence is the point
- **Result on real frames: near-zero.** The test is measuring the wrong amplitude

FIGURE / DIAGRAM. left: the measured recovery matrix as a heat map; right: a two-panel explanation of why – the variance plane versus the actual residual structure of a co-add.





amplitude = $\rho\,\overline{\sigma}$ is far below the real residual
⇒ a "20σ" fake source is invisible. The test is wrong, not the pipeline.

SPOKEN SCRIPT.

A detection chain is only as good as its completeness curve, so stage six plants fake sources and counts how many come back. The grid is signal-to-noise from three to twenty, proper motion from one to ten arcseconds per year, twenty trials per cell, and a trial counts if the right kind of detection lands within two pixels of the injection.

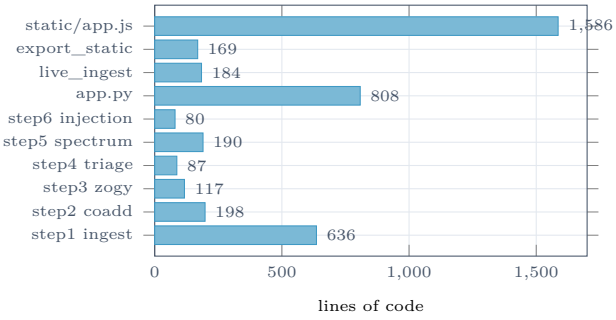
And here is the uncomfortable result, which I am showing you rather than hiding. On real frames, almost nothing is recovered: for Barnard's Star, zero out of twenty-four cells; the best field reaches three in ten only at the very brightest end. The reason is on the right. The amplitude I inject is tied to the mean of the variance plane, which is flat, while the real residual in a co-added frame is dominated by PSF mismatch, which is not in the variance plane at all. So a fake source injected at "twenty sigma" is a barely visible bump on top of the real residual. The test is measuring the wrong thing – and until I fix it, I cannot quote a completeness limit, which is why the next slides say unexplained rather than new.

Implementation I: the code, and what changed

ON THE SLIDE.

- Six stage scripts (1 318 lines) + service (808) + front end (1 586) = 6 154 lines
- Stages are importable and runnable: `python step3\_zogy.py` works on synthetic data
- Five revisions turned a demo into a system (R1–R5) – each one was a bug with a lesson
- Everything deterministic; one metadata file is the contract between stages

FIGURE / DIAGRAM. a horizontal bar chart of lines per module, next to a short code excerpt showing the "undefined noise" guard that stops the service printing a meaningless depth.



R1 – undefined noise must not print as a number

```
def usable(sig):
  return sig if isfinite(sig) and 1e-4 < sig < 1e3 else None
```

Betelgeuse saturates, the inverse-variance denominator collapses, and the old code printed a depth of 1.6×10^6. Now it reports *no estimate* and says why.

the other four

R2 epoch grouping is data-driven, not half-and-half
R3 arbitrary fields live in a RAM registry, not on disk
R4 a static build disables what it cannot do – and says so
R5 the interface explains itself (a three-tab guide)

SPOKEN SCRIPT.

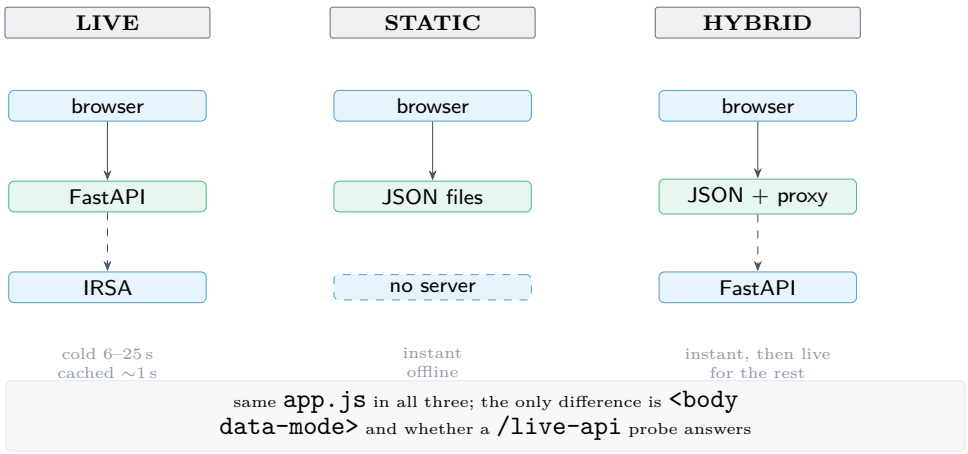
On the engineering. The pipeline is six scripts totalling about thirteen hundred lines, one service of eight hundred, and a front end of sixteen hundred – six thousand one hundred and fifty-four in total, which is small enough that a reviewer can read all of it in an afternoon. Each stage has its own entry point, so I can re-run one stage on cached data when something looks wrong.
Five revisions matter enough to name. The most instructive is R1, on the right: Betelgeuse saturates the detector, almost every pixel is flagged, the inverse-variance denominator collapses, and the old code happily reported a five-sigma depth of one-point-six million. That is worse than reporting nothing, because it looks like a measurement. The fix is four lines: anything outside a physically possible range is "no estimate", with a note saying why. The other four are the same flavour of honesty – data-driven epoch grouping, a RAM registry for arbitrary fields, a static build that disables what it cannot do and says so, and an interface that explains itself.

Implementation II: one result set, three deployments

ON THE SLIDE.

- **Live**: FastAPI, five read-only endpoints, 30-minute in-RAM cache, nothing written
- **Static**: every response pre-rendered to JSON – 28 MB, works with no server
- **Hybrid**: static page + a same-origin `/live-api` proxy, discovered at load
- A static build disables "live from IRSA" and says why instead of failing silently

FIGURE / DIAGRAM. three columns for the three shapes, with the request path drawn for each, plus a row showing what each costs in latency and in what it can do.



SPOKEN SCRIPT.

The last piece of the engineering is how the result is served, and I deliberately built it in three shapes because each answers a different question. Live: a FastAPI service with five read-only endpoints, a thirty-minute in-RAM cache, and nothing written to disk – thirty to sixty seconds for a cold field, about a second when cached. Static: a script renders every response for every field to JSON, which is twenty-eight megabytes and works from a USB stick, no server at all. Hybrid: a static page with a same-origin proxy to a Python host, which is how a static host like Netlify can still offer the live features.

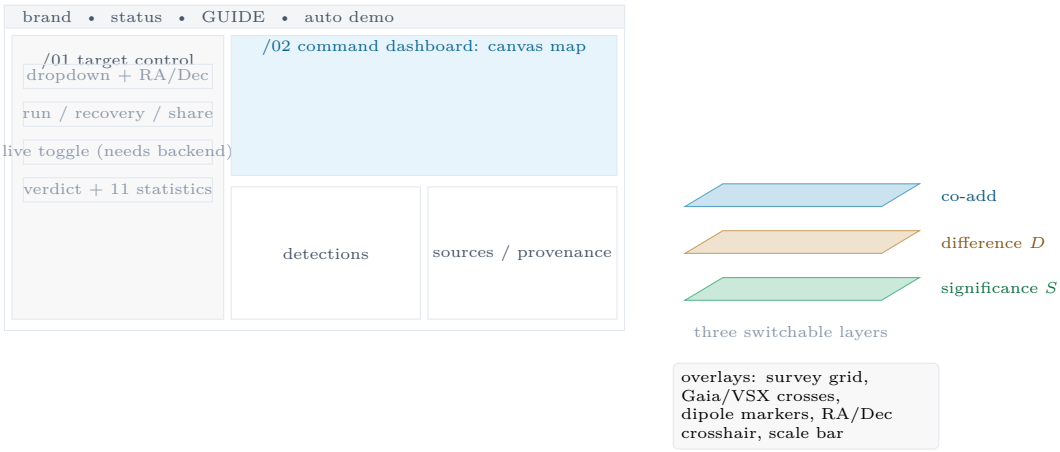
One consequence is worth highlighting because it is a design principle rather than a detail. On the pure static deployment, live fetching physically cannot work – so the page disables that control, relabels it 'live from IRSA, needs a backend', and prints the reason underneath. If a proxy appears later, a single probe at load turns it back on. A toggle that silently does nothing is worse than a toggle that is switched off.

Implementation III: the front end

ON THE SLIDE.

- 1 586 lines of vanilla JavaScript; no framework, no build step, no tracker
- Canvas engine: three layers (co-add, difference, significance) + catalogue and survey overlays
- Everything is one JSON per field: map, detections, objects, spectrum, provenance
- A three-tab guide written for someone who has never seen an astronomical image

FIGURE / DIAGRAM. a wireframe of the actual page with the panels labelled, and the canvas layer stack drawn beside it.



SPOKEN SCRIPT.

The front end is sixteen hundred lines of plain JavaScript with no framework and no build step, which is why it will still run in five years. It is a small canvas engine with three switchable layers – the co-add, the difference image, and the significance map – plus overlays for the survey grid, the catalogue positions, the dipole markers and a readout of whatever is under the cursor. On the left, target control: pick a field, run, look at the verdict and eleven statistics, or ask for a recovery grid.

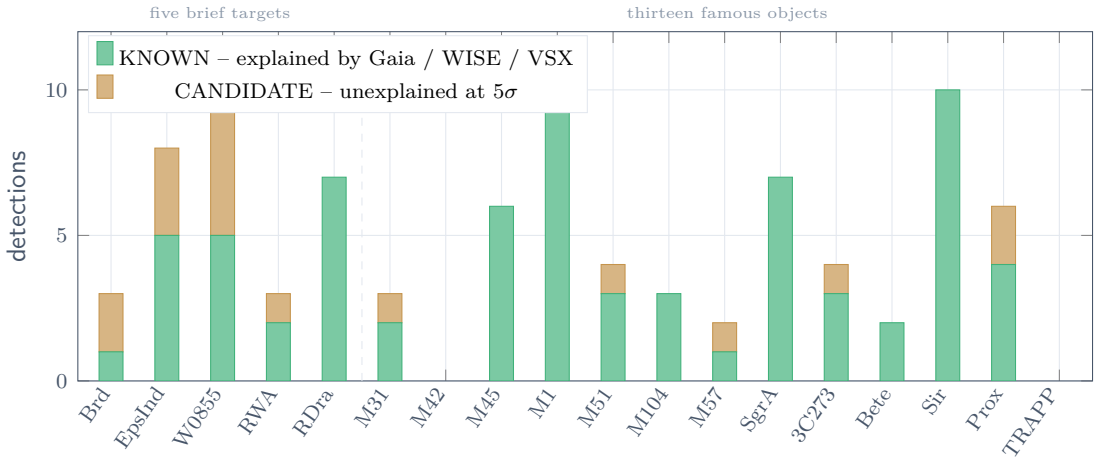
The thing I want to point at is the Guide button in the top bar. It opens three tabs written for someone who has never looked at an astronomical image: how it works, how to use this page click by click, and a glossary of the ten words that matter. It is plain HTML inside the page, so it ships in the static build and needs no backend – and it states the backend limitation in the same place as everything else.

Results I: eighteen fields, eighty-eight detections

ON THE SLIDE.

- 88 detections over 18 fields: **71 KNOWN** (81%) and **17 CANDIDATE**
- Detection count is not discovery count: M1, Sirius and R Draconis are 100% explained
- The amber bars – WISE 0855 (5), Epsilon Indi B (3), Barnard's Star (2) – are the interesting ones
- Four fields produce nothing usable, and they are on the next slide

FIGURE / DIAGRAM. stacked bars for all eighteen fields, grouped visually into the five brief targets and the thirteen famous objects, with the totals annotated.



SPOKEN SCRIPT.

Here is the whole result in one chart. Eighteen bars, one per field, green for detections the catalogues already explain and amber for the ones they do not. Two things to notice.

First, the distribution of detections is not the distribution of discoveries. The Crab, Sirius and R Draconis all look busy – ten, ten and seven detections – and every single one is explained, which is the correct outcome: these are famous, well-catalogued objects. The interesting bars are the amber ones: five in WISE 0855, three in Epsilon Indi B, two in Barnard's Star.

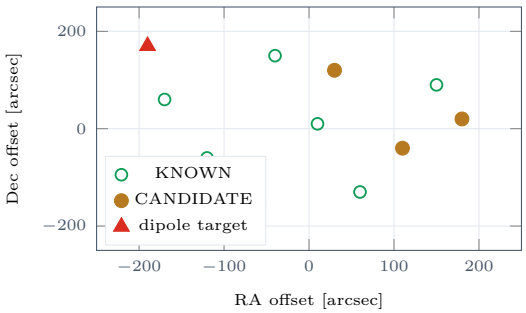
Second, look at the sample itself. Five brief targets on the left, thirteen famous objects on the right – galaxies, nebulae, a quasar, the Galactic Centre, Betelgeuse, Sirius. Every one went through the identical code path: one hundred and twenty-eight cutouts, eighty-eight detections, eighty-one percent explained by catalogues. That is the strongest evidence I can offer that the pipeline is not tuned to one kind of object.

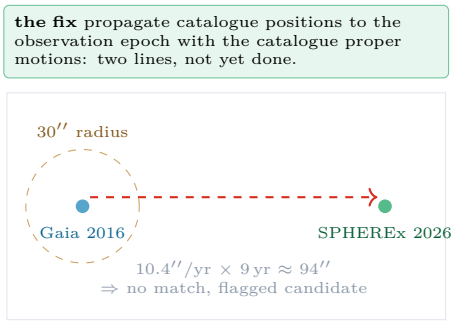
Results II: what the seventeen candidates actually are

ON THE SLIDE.

- Most are positions where a $> 5\sigma$ change has no catalogue source within $30''$
- Five in WISE 0855 – the field with the fewest catalogue sources per unit noise
- Barnard's Star's 56.8σ dipole is *itself* a mislabel: epoch mismatch, not a discovery
- So: "candidate" means **unexplained**, not **new** – the distinction is the result

FIGURE / DIAGRAM. left: a scatter of the detections in one field (RA/Dec), showing the dipole pair and the candidates; right: a cartoon of the epoch-propagation problem with a Gaia position from 2016 and a SPHEREx position from 2026.





SPOKEN SCRIPT.

What are the seventeen? I want to be careful here, because this is where a project like this usually gets oversold. The scatter on the left is a typical field: hollow circles are detections the catalogues explain, filled dots are the unexplained ones, and the triangle is the target itself.

Most of the seventeen are simply positions where a five-sigma change has no catalogue source within thirty arcseconds. Five of them are in WISE 0855, which is the field with the fewest catalogue sources per unit noise – exactly what you would expect if the candidate rate tracks catalogue density rather than astrophysics. And the single most significant detection in the whole sample, Barnard's Star at fifty-seven sigma, is itself a mislabel, for the epoch reason on the right: Gaia's position is from 2016, my image is from 2026, the star has moved ninety-four arcseconds, and nothing matches.

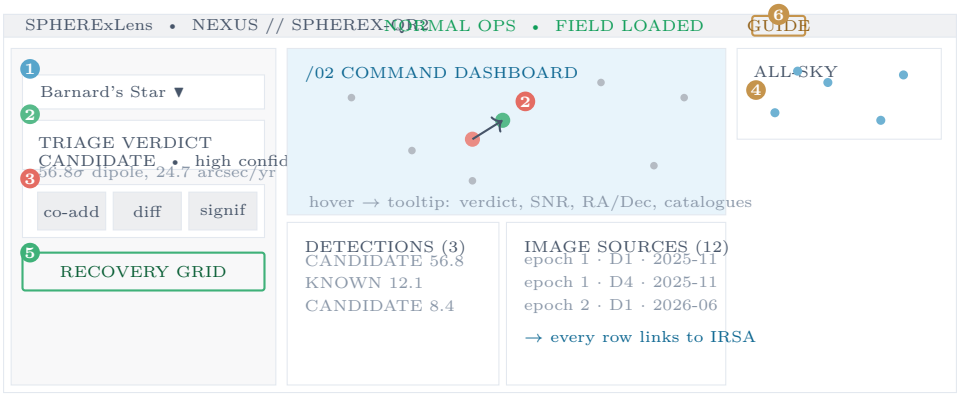
So the result I am actually claiming is narrow: seventeen positions where a five-sigma change is unexplained by three catalogues. Not seventeen new objects. The word on the slide is unexplained.

Live demonstration

ON THE SLIDE.

- **Open** `site/index.html` (already loaded in a second tab, zoom 100%)
- **1** Pick *Barnard's Star* – point at the dipole, hover it
- **2** Click it: verdict panel, SNR, proper motion, catalogue matches
- **3** Switch layers: co-add → difference → significance
- **4** All-sky view → pick *Crab Nebula* (10 detections, all KNOWN)
- **5** *Recovery Grid* on the current field; read the matrix aloud
- **6** *Guide* → tab two, to show the user-facing explanation

FIGURE / DIAGRAM. an annotated map of the screen: the six numbered callouts sit on the exact controls used, so the presenter never hunts for a button on stage.



SPOKEN SCRIPT.

I would like to show this working rather than describe it, so I have the site open in the second tab. This is the static version – no server – so everything you are about to see is pre-computed JSON being drawn by the browser.

First, Barnard's Star, which is the demo I would recommend to anyone. The map shows the significance of the difference between two visits. Point at that bright pair in the middle – which is a dipole, the star's old position in blue, its new one in green – and the tooltip tells you the verdict, the significance, and the catalogues that match. Click it and the verdict panel opens: this is the fifty-seven-sigma detection, a proper motion around twenty-five arcseconds a year, and no catalogue match, for the epoch reason I explained.

Now the layer switch: co-add, then difference, then significance – that is the whole chain, visible one step at a time. Over to the all-sky view, where every field is a thumbnail; I will pick the Crab, which gives ten detections and all ten explained, because that is the contrast I want you to see. Finally the recovery grid for the field, which is the completeness matrix, and the Guide tab, which is what a visitor sees before anyone explains anything to them.

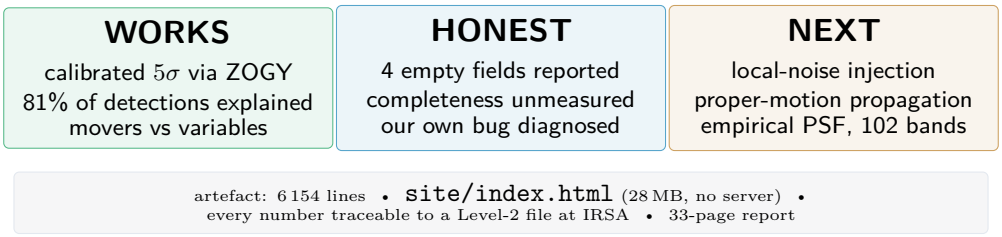
If the laptop misbehaves, the fallback is the same content in the report – figures three, six and seven – and I will move on rather than debug in front of you.

Conclusion, contributions, and questions

ON THE SLIDE.

- **Works**: ZOGY on masked co-adds gives a calibrated 5σ map; morphology splits movers from variables; triage explains 81% of detections
- **Honest**: four fields return nothing, completeness is unmeasured, one candidate is my own bug – all reported
- **Next**: local-noise injection, proper-motion propagation, empirical PSF, all 102 bands
- Artefact: six stages, the derived JSON, a static site, and a guide for non-specialists

FIGURE / DIAGRAM. three pillars (works / honest / next) with the artefact listed beside them, and a compact reference strip at the bottom.



references: Alard & Lupton 1998 • Bramich 2008 • Zackay, Ofek & Gal-Yam 2016 • Hu et al. 2022 (SFFT) • ZTF/DES pipelines and the broker literature (ANTARES, ALeRCE, Lasair) • Gaia DR3; AllWISE; AAVSO VSX • SPHEREx mission and IRSA documentation

SPOKEN SCRIPT.

To close, three statements and then I will take questions.
It works: proper subtraction on masked co-adds gives a significance map whose threshold means something, the morphology separates movers from variables where the displacement exceeds the PSF, and catalogue triage explains four fifths of what the pipeline finds.
It is honest: four fields came back empty and the deck says so, the completeness is still unmeasured rather than invented, and the single most significant detection in the sample is a bug I found in my own triage and have described rather than hidden.
And the next three changes are small and each one converts a caveat into a number: inject at the image level with local noise, propagate catalogue proper motions before matching, and use an empirical PSF per detector. Everything is in the repository – the six stages, the derived JSON, the static site, and a guide written for someone who has never seen an astronomical image. Thank you; I am happy to take questions.

Backup: five questions, and the pre-flight checklist

Q&A PREPARATION.

"Why only 2–3 bands and not 102?" Cost: 102 cutouts per epoch per field is roughly 500 MB and 70 minutes of archive time *per field*. Subtraction does not need the extra bands; the spectral indices do. It is the first item on the next-steps list.

"Is 30″ not far too generous a match radius?" For astrometry yes; for vetoing it is deliberate. A loose radius biases towards KNOWN, which is the conservative direction – the alternative inflates the candidate list and flatters the result.

"What is the real false-positive rate?" Predicted 0.018 pixels per frame at 5σ; observed detections are tens of sigma above that floor, so the map is not noise-dominated. I do not quote a measured rate because completeness is unmeasured.

"Could this scale to the whole sky?" The CPU could: 6 s per field times 1.8 million 9′ fields is about 125 days on one core. The archive is the bottleneck, which is why the design caches aggressively and pre-renders.

"Why ZOGY rather than SFFT?" ZOGY is closed form and its null distribution is analytic, which is what makes the false-positive budget explicit – the argument on slide 12. SFFT would be better for an undersampled space PSF and is listed as future work.

PRE-FLIGHT CHECKLIST (five minutes before).

- Two browser tabs: `site/index.html` (static, offline-safe) and, if the network is good, the local API on `localhost:8000` for the live fetch.
- Zoom 100%, one field already loaded (Barnard's Star), so the demo does not start on a spinner.
- Screen recording of the demo as a fallback file on the desktop.
- This document open at page 23 (the demo map) so the callouts match what is on screen.
- Numbers to have on the tip of the tongue: 18 fields, 128 cutouts, 88 detections, 71 explained, 17 unexplained, 4 empty fields, 0.018 false pixels per frame.